



Two uniform smooth spheres A and B of equal radii have masses m and $\frac{1}{2}m$ respectively. The two spheres are moving on a horizontal surface when they collide. Immediately before the collision, sphere A is travelling with speed u and its direction of motion makes an angle α with the line of centres. Sphere B is travelling with speed $2u$ and its direction of motion makes an angle β with the line of centres (see diagram). The coefficient of restitution between the spheres is $\frac{5}{8}$ and $\alpha + \beta = 90^\circ$.

- (a) Find the component of the velocity of B parallel to the line of centres after the collision, giving your answer in terms of u and α . [4]

[illegible]

The direction of motion of B after the collision is parallel to the direction of motion of A before the collision.

(b) Find the value of $\tan \alpha$. [5]

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